

Collaborative Diffusion: Boosting Designerly Co-Creation with Generative AI

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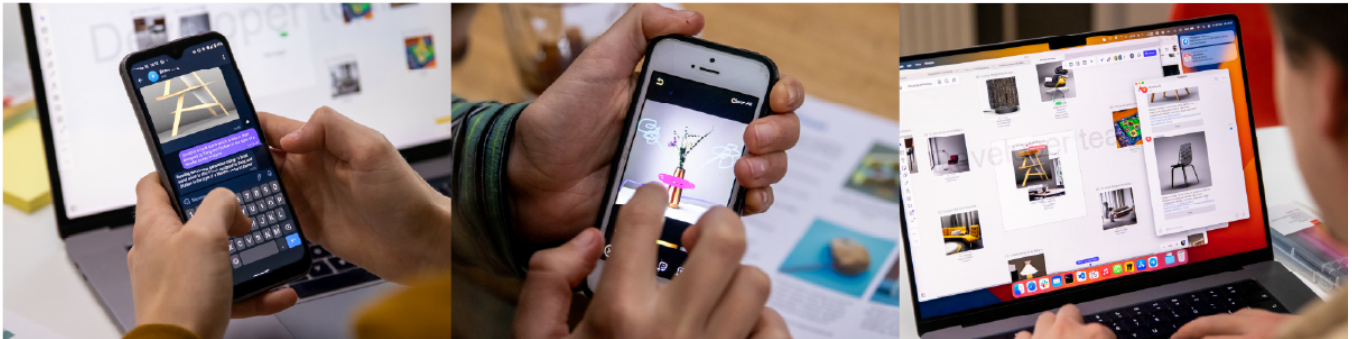


Figure 1: Visual overview of BrainFax-supported co-creation activities. From left to right, generating images in Telegram chat using text prompts, sketching over images in Telegram to guide AI adjustments, organizing and annotating images in Miro.

ABSTRACT

Visual communication in collaborative design provides common ground and a tangible trace of thoughts. Yet, it is often underused due to a lack of visualization skills and time needed to sketch and detail. Current visual co-creation tools can support this with existing images, whereas generative AI can produce entirely new images from textual prompts, allowing for highly specific, personalized results. In this paper, we explore how AI image generation can be used to enhance inspiration and communication throughout collaborative design. We introduce BrainFax, a tool that facilitates generating, editing, and sharing images with AI through a chat bot and online whiteboard. Through co-creation with designers in the field, we found that AI image generation can boost designerly co-creation, yet needs careful embedding into a workflow to leverage inspirational and communicative creations. We close with a critical perspective on the implications of this technology for design and discuss limitations and future work.

CCS CONCEPTS

• **Human-centered computing** → **Systems and tools for interaction design; Interaction design process and methods; Interactive systems and tools.**

KEYWORDS

Generative AI, AI Image Generation, Co-Creation Tool, Design Process, Human-AI Collaboration

ACM Reference Format:

Mathias Peter Verheijden and Mathias Funk. 2023. Collaborative Diffusion: Boosting Designerly Co-Creation with Generative AI. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (CHI EA '23)*, April 23–28, 2023, Hamburg, Germany. ACM, New York, NY, USA, 8 pages. <https://doi.org/10.1145/3544549.3585680>

1 INTRODUCTION

In collaborative design sessions, we share, document, and combine ideas to come up with new innovative concepts, but a lack of visual representation of these ideas often results in long explanations or hand gestures, trying to compensate for the lack of common ground. This limitation to express oneself can be a challenge in group performance [22]. Visual information can bridge the gaps in our communication by providing common ground, persistence, and stability by leaving a visible and tangible result of our expressions, making fluent thoughts concrete [3, 27]. Visuals represent information at high bandwidth and at-a-glance without relying on a specific chronological starting point, unlike most verbal summaries [27]. According to Snyder [28], the most common reasons for using visuals during collaboration are to *clarify*, i.e., providing additional visual information to fill communicative gaps, to *inventory*, gathering and listing (examples of) what already exists, and to *show*, visually describing a tangible object instead of explaining it. Using visuals to communicate increases the participants' involvement [3] and improves collaboration speed, creativity, and effectiveness [4]. This should not come as a surprise. We are so focused on visual material in our daily lives that 75% of our sensory neurons are visual neurons [4]. Still, despite the benefits of visually

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CHI EA '23, April 23–28, 2023, Hamburg, Germany

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ACM ISBN 978-1-4503-9422-2/23/04.

<https://doi.org/10.1145/3544549.3585680>

supported collaboration in design, many of us (need to) rely on verbal or written communication. Not only can it be time intensive, but we are quickly discouraged from working visually due to a lack of skills or practice [4]. In complex conceptual co-creation sessions, this limits how we can express ourselves in terms of depth, speed, and fidelity.

In this paper, we report on exploratory research into supporting creative collaboration with generative AI. As a means to research with participants in the field we developed a visual co-creation tool: BrainFax. Through an integration with a Telegram [31] chat bot, the online collaboration and whiteboard platform Miro [16], and the Stable Diffusion API [24], BrainFax can be used to generate and edit images using text prompts and simple sketches, as well as share, organize, and annotate these images with others. In the following, we first introduce related work and provide an overview of BrainFax and its functionality. We then describe the approach and results of a pilot test of BrainFax with experts. Through initial insights from a follow-up co-creation, we provide design guidelines in the form of a workflow that shows how to integrate generative AI into creative collaborative practices. We discuss implications of our findings for AI-mediated creativity, reflect on limitations, and conclude with an outlook on future work.

2 RELATED WORK

2.1 Visual Co-Creation and Creativity Support Tools

Several projects have been developed to enhance visualization through computational means and support a more technology-mediated creative process. A first example is TalkTraces [6], a tool that captures verbal discussions and visualizes them in word clouds in real time. Although not strictly visual, it created a tangible trace of verbal expressions to facilitate further debate. However, as it still relied on text, users found it challenging to recall topics at-a-glance. Idea Expander [33] intelligently shows pictures from a labeled database that are related to the current conversation. The visual stimuli were meant to support idea exchange and idea expansion, by triggering participants' associative memory. This is called the "cognitive stimulation effect." Study participants were able to generate 20% more ideas using Idea Expander. Idea Wall [26] also analyzes the current conversation, but searches for relevant images on Google Images. Speech recognition errors sometimes resulted in a mismatch between conversation and images. However, Shi et al. [26] found that users could sometimes draw inspiration from these mismatched results. FashionQ [11] uses AI to find relevant inspirational images. Based on source images, it provides up to 45 images of other outfits that fit within similar fashion styles. Style combinations can also be requested, for which FashionQ will provide outfits that contain attributes from both styles. ImageSense [14] is a digital whiteboard for collaborative idea generation using images. Users can search for specific images, find them randomly in a preselected cascade, or get relevant images recommended by AI. They found that users could communicate ideas better, felt more creative, and explored more alternatives. The image-based tools discussed in [11, 14, 26, 33] used databases of existing images, which sometimes had to be curated and labeled by hand, taking extensive time and resources [11, 33]. The system's applicability therefore

also depends on the finite source material. They are therefore unable to provide new content that is not already present, limiting the effectiveness of these systems [11, 14, 26, 33]. Users of ImageSense and FashionQ also indicated that they would like more fine-grained control over the automatically suggested images [11, 14].

2.2 Diffusion Models: How does Generative AI work?

AI image diffusion models are trained on millions, sometimes billions of pictures containing a wide variety of subjects. Compared to previous work that uses manually curated databases of existing images [11, 14, 26, 33], their applicability is much less limited by the source material. This is because instead of providing source material, new images that go beyond the original data set can be generated. Because of these characteristics, Dang et al. [9] already proposed applications for generative AI in creative industries such as fashion and interior design, albeit with the use of older Generative Adversarial Network (GAN) [7] technology. Despite slightly different approaches, the overall concept is the same for all Latent Diffusion Models (LDM): first, deep learning is used to find patterns in image data. In a "latent space," these patterns begin to form parameters that describe the characteristics of depicted objects, e.g., an object's "roundness" [25, 32]. Done for millions of pictures, clusters emerge that describe certain objects [19, 32]. For example, what makes a bike look like a bike? Second, with this training data, a model is trained to predict what is noise in an image and what is not. By starting with pure noise and letting the model predict all the noise in the image, a final image could theoretically be generated by subtracting the identified noise. However, these models are not flawless, which is why a little noise is removed step by step so that each step can correct and improve from the previous one [25]. Finally, a model is used to convert a text prompt into embeddings, which help determine which words influence each other. For example, "a man with blue eyes" should result a visual of blue eyes, not a blue man [21]. These embeddings are used to steer the noise predictor in the direction of the prompt. These separate parts are used sequentially, in which the output from one model is used as input for the next, translating your text prompt to an image.

2.3 AI Image Generation Tools

With the introduction of DALL-E 2 [18, 23] and MidJourney [15], high-quality AI image generation became available to the public. However, with the recent release of Stable Diffusion (SD) [17], available open-source, the technology became accessible to design and development of new tools and applications—according to Rombach et al. [25, p. 1] the "*democratization of high-resolution image synthesis*". SD is also available as cloud-based tools like Dream Studio [30] or Replicate [24], as well as local software packages like Diffusion-Bee [10], Stable-Diffusion-Web-UI [1], or NoliBox [5]. For design, the downside of these applications is that they are single-user and result-oriented, focused on fine-tuning desired image output. That is, these tools target different design activities and outcomes: "*Users may share the gems they discover, but my guess is that 99 percent of the 20 million images currently generated each day will only ever be viewed by a single human*" [13]. In the context of this paper, we are

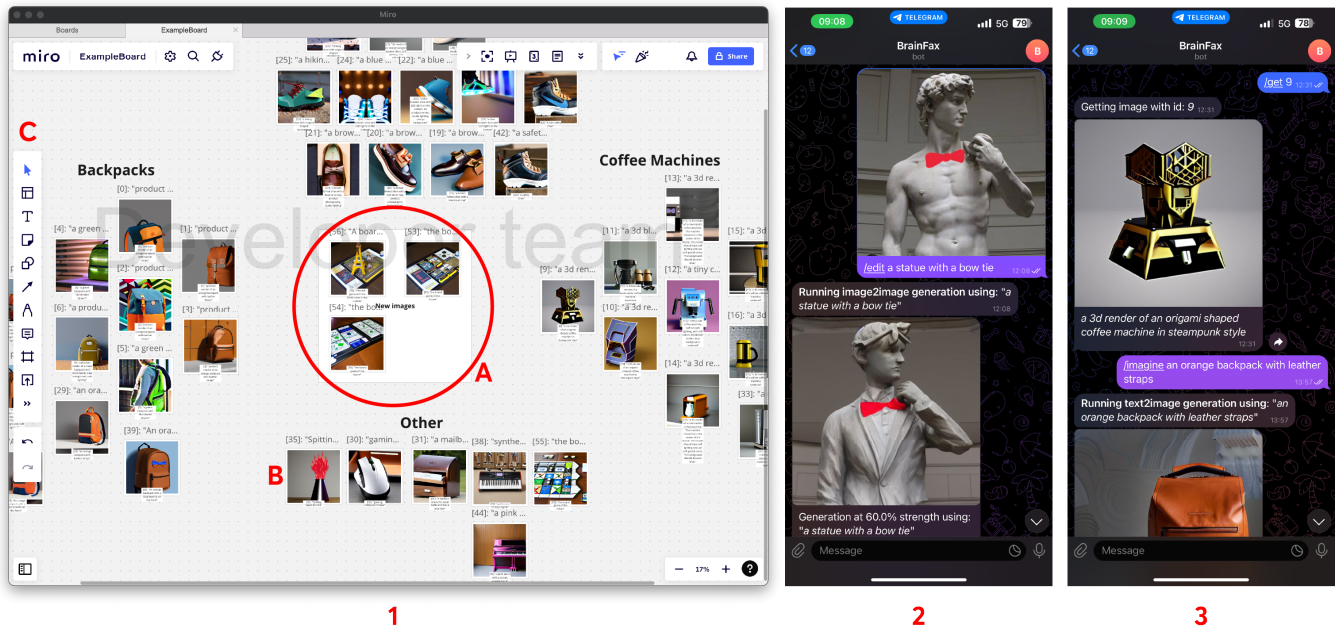


Figure 2: Overview of Miro (1) and Telegram (2 and 3) interfaces. The Miro interface features a 'New Images' section (A), where all newly saved images are placed. These can then be freely distributed across the board. Images on the board (B) display their ID number and the corresponding generation prompt. Using Miro's built-in toolbar (C), images can be further organized and annotated. Images in Telegram are generated using `/imagine [prompt]` (3), which returns four images. Existing images can be retrieved using `/get [id]` (3). These images can be edited by adding a rough sketch using Telegram's built-in sketching tools and using `/edit [updated prompt]` (2). This will also return four new images.

more interested in exploring how humans and AI can work together in a multi-user and process-oriented manner of using generative AI: not the engineering of prompts, but a fluent, fast-paced generative loop with the potential to inspire and help communicate, in the sense of "... the challenge is not building the AI agent itself, but rather engineering the human-machine teams that leverage it" [8, p. 24].

3 BRAINFAX: AI-POWERED VISUAL CO-CREATION

We designed BrainFax to be used during design sessions that involve multiple participants, such as designers or other stakeholders. To make use of the benefits of visual communication, without relying on the visualization skills of the participants themselves, BrainFax provides a set of features to create images using Stable Diffusion. Using BrainFax, users can perform image generation, adaptation (editing), and curation (sharing and organizing), to support the visual co-creation process.

3.1 Interaction

Image generation is performed within the Telegram chat interface in two forms: (1) text-to-image, using a textual prompt, and (2) image-to-image, using a textual prompt as well as an input image. This can be seen in Figure 2. Using `/imagine[prompt]`, four different images are generated based on the textual input. For image adaptation, any picture in the chat can be selected and edited with Telegram's built-in sketching features, e.g., drawing over a part of an image.

By sending this sketched-over image with `/edit[prompt]` in its caption, four new images are returned, based on the image and the refined prompt. Because SD reinterprets the entire image based on the prompt, the input image and sketch are combined into a coherent image. Therefore, very little sketching skills are required to make very quick targeted adjustments. New images in the chat can be saved using the `save` button under each image. They are then displayed on the linked Miro board where they can be curated using Miro's built-in organization and annotation tools.

3.2 Collaboration

By sharing images to Miro, material can be rearranged, grouped, or annotated, which are usually essential activities in collaborative co-creation sessions. This provides a visible and tangible result of the user's expressions, bridging communicative gaps by providing common ground, persistence, and stability—a central place for reflecting and discussing each other's contributions. To further strengthen the collaborative nature of BrainFax, images made by others can also be retrieved in one's individual Telegram chat, as can be seen in Figure 2. Using the command `/get[id]` with the image displayed on the Miro board, users can use sketching and image-to-image features to iterate on each other's work, providing an additional layer of interpretation and reflection. Although image generation is not instant, the speed of the Replicate SD instance is high enough for a lively collaboration session with 5-7 participants (see Tool Evaluation).

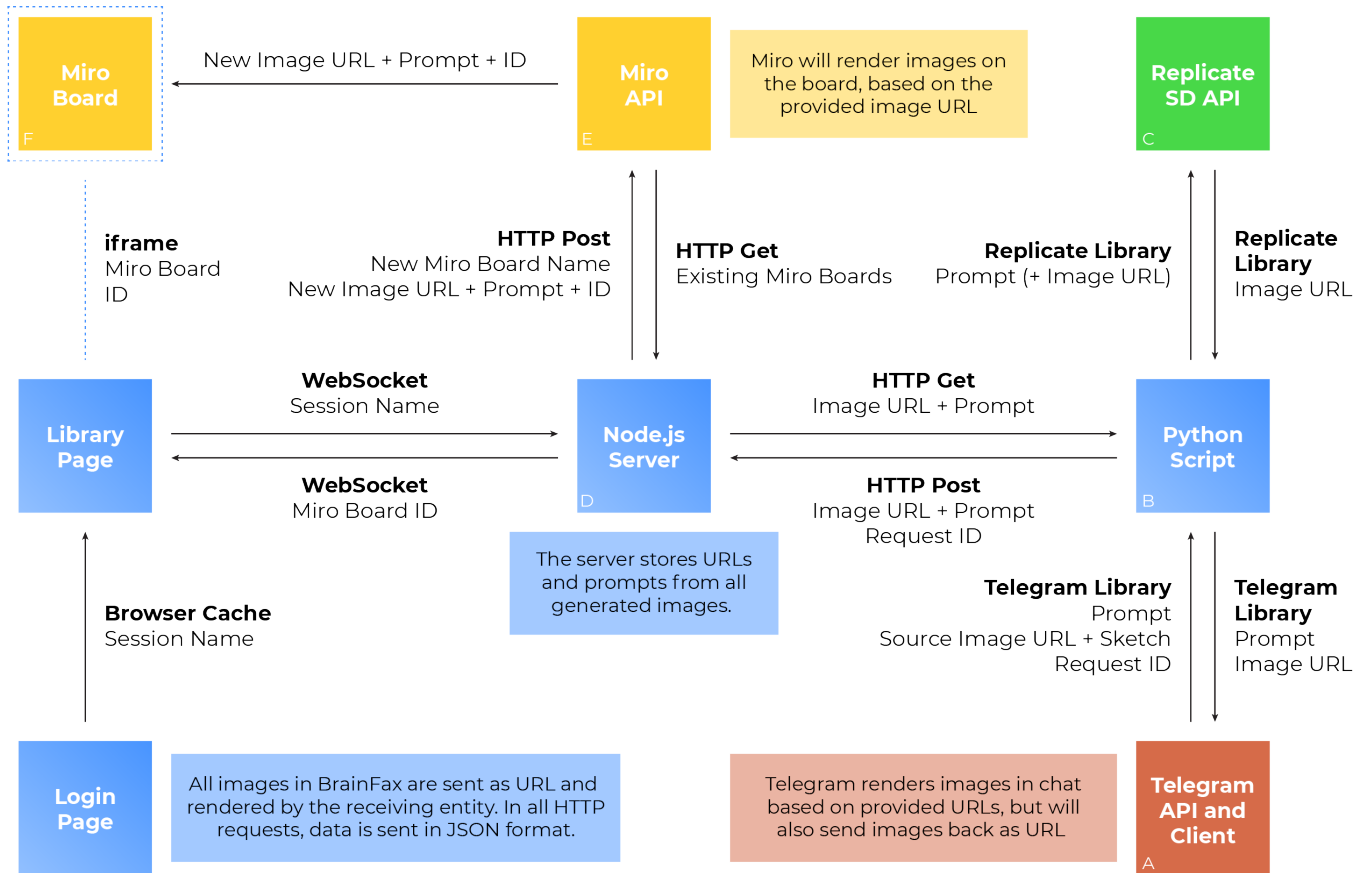


Figure 3: Technical overview of BrainFax Prototype. The individual user interface, Telegram (A), is used to generate and edit images through textual commands `/imagine [prompt]` and `/edit [prompt]`. Through a Python script (B), Telegram can communicate back and forth with the Replicate API (C). This API accepts, next to other technical settings, a textual prompt and input image which return four image generations as URLs. Telegram automatically displays image URLs as images in the chat. When an image is saved in Telegram, the Python script sends the corresponding URL and prompt to the server (D). The server stores all saved image URLs and corresponding prompts. With every save request, prompt and URL are sent to the Miro API (E), which displays it as an image and caption on the Miro board (F). Users can collaboratively rearrange and annotate images using all of Miro's features. Finally, when requesting images made by others with `/get [id]`, due to Miro API constraints, the corresponding URL and prompt are not provided directly by Miro but by server storage and sent to Telegram chat.

3.3 System overview

The BrainFax system is made up of two different main sections: (1) the image generation system, and (2) the image sharing system. A technical overview is provided in Figure 3. The first section consists of Telegram API, Replicate Stable Diffusion (SD) API, and a connecting Python script. Telegram provides IDs for creating bots, so that messages sent to that bot can be processed using custom scripting. By using specific inline chat commands in Telegram and providing an image prompt, specific functions in the script are triggered that call the SD API to generate images. Responses from the API containing images are then correctly formatted and returned to the Telegram client using the Telegram API. The second component consists of the Miro API and a Node.js server that forms the bridge with the first component and Miro. The server stores all images and corresponding prompts in a local database. Each

time a new image is saved in Telegram, the Python script sends the image and prompt to the server, which forwards it to the Miro API. This API will then place the image on the Miro board with the corresponding prompt in its caption. Images generated by the SD API are delivered in URL format and sent to both Telegram and Miro. Images are therefore only rendered in Telegram or Miro.

4 TOOL EVALUATION: CO-CREATION SESSIONS WITH BRAINFAX

While BrainFax was designed to offer robust features to facilitate visual communication, that does not mean that every designer knows what to do with it. Snyder et al. [29] used sketches as speculative means of reflection and inspiration to elicit stakeholder discussions. They found that many participants interpreted the sketches as concrete proposals for products to be implemented. As designers, we

are trained to see the individual elements and opportunities in the whole. But how will the much higher fidelity AI-generated images be seen and used in co-creation?

4.1 Testing session

To explore how BrainFax is—and should be—used in the design process, two separate testing and co-creation sessions were organized at a commercial product design agency to gather perspectives from designers in the field. The testing session was conducted with five participants, two of whom also participated in a semi-structured interview about their experiences. This interview was not conducted in English, so quotes were transcribed, translated, and analyzed through thematic analysis. The quotes are not attributed to individual designers. Later, the co-creation session was conducted with nine participants, four of whom also participated in testing. Participants, aged between 20 and 32, had a background in industrial design or mechatronics. The objective of the testing session was to explore how BrainFax would be used in a real-world scenario. Therefore, it was used for one of the design agency's current design projects. The session started with a quick introduction and demonstration of BrainFax. In addition, a prompting "cheat sheet" was provided with tips and tricks for writing effective image prompts, including keywords for specific styles. No further instructions were given on when and how to use BrainFax to capture the designers' natural use and reactions.

The first fifteen minutes of the session were spent silent, with all participants continuously generating images. Although images were shared, they were not discussed by others: *"it is difficult to break out of the cycle and go back to your normal brainstorm mindset"*. Despite some attempts to initiate discussion, the focus remained on the individual generation. After the participants finished *"playing around"*, the conversation slowly started. Some results were shared and discussed, but the lack of a common goal for the session made participants unsure of what to do with the generated material. It was clear that the novelty effect had dominated the session. The image generation itself was also found to be quite challenging; mainly because participants tried to visualize specific functionality, like *"a [object] that can [functionality]"*. They noted that *"you are essentially trying to visualize a feature, instead of finding a certain shape for example"*. Because functionality is very abstract and can be achieved in hundreds of ways, prompts like these are very difficult to generate for AI: *"You are essentially fighting with the AI to let it make something that you already have in mind"*. This also strongly indicates that, as expected, the AI was seen as a tool for generating final designs. Because participants have therefore already converged a lot, their frame of acceptance becomes very small, leaving little room for inspiration. We call this the 'crooked bow tie' effect, as explained in Figure 4, which ideally features equally shaped sides.

Despite their difficulties, participants were enthusiastic about the potential of BrainFax, noting that it was perhaps best for diverging activities: *"it has inherent characteristics that are very nice for diverging"*, which they also found in practice: *"When I went to more simple prompts it worked much better. It was perhaps not exactly what I was looking for, but it allowed me to get much more insights*

out of [the images]". What they liked most was how BrainFax facilitated collaboration and personalisation through sketching: *"I think the sketching and collaborative part is very cool"* and: *"What was nice was the /get function so that you can add to [other work]"*. Mostly though, participants emphasized that they would like to have some guidelines or methodology to work with BrainFax: *"I think it [BrainFax] would be very valuable if there are guidelines for how to integrate it,"* and *"I think it should be part of a methodology, so you know when to bring it [BrainFax] to the fore- and background."*

4.2 Co-Creation session

The findings from the testing session were used to develop and test a creative session format with a focus on integrating generative AI with a focus on collaboration. This was done in a co-creation session with designers from the same design agency. Similar to the testing session, a quick demonstration and cheat sheet were provided. The first 15 minutes were spent on the assignment *"what if Bang and Olufsen made furniture"* to get familiar with BrainFax's capabilities. The initial findings of the testing session were then discussed. This resulted in three focus points for the co-creation session: (1) how to balance individual and group activities, (2) how to shift image generation between fore- and background, and (3) how to get value from the generated images. These topics were discussed in three subgroups, after which a plenary discussion followed.

Participants found it challenging to direct the generation process in terms of scope and specific prompts. They suggested *"having a pre-made library [...] in which you can look up keywords to use [for prompting]"* for example, so that everyone can work towards a common goal, i.e., establishing what and why we are generating. They also found that the image generation process goes too quickly. Therefore, *"...it is almost impossible to react to each other's images"* and *"More in-between sharing"* is necessary. Therefore, they suggested to *"Limit the amount of prompts you can have"* and make generation sessions no longer than 10 minutes at a time. This way, image generation is deliberately shifted to the background, forcing the group to reflect and discuss the present material. In terms of reflection, participants proposed *"Voting on ideas that are a cool direction"* to create a set of images deemed desirable. This can be used for analysis, i.e., deriving a set of overlapping characteristics defining the team's design preferences. In summary, three phases were proposed to; (1) establish a goal and direction, (2) generate new images, and (3) analyze the results. Participants suggested performing phases one and three collectively, to establish a common starting point and facilitate discussion, but to perform phase two individually, to give room to personal interpretation, exploration, and perspectives.

5 DISCUSSION

At the time of writing, there are many online discussions on *"how AI will take over our jobs"*, and how it will be a *"danger to the (creative) industry"*. These perspectives are grounded in a vast overestimation and misinterpretation of AI capabilities; understandable, plausible, but wrong. When prompting MidJourney for *"an orange leather backpack"*, it produces a very high-quality flawless image you could easily imagine in stores. These refined visualizations tend to be accepted 'as it is', whereas rough versions such as sketches tend to

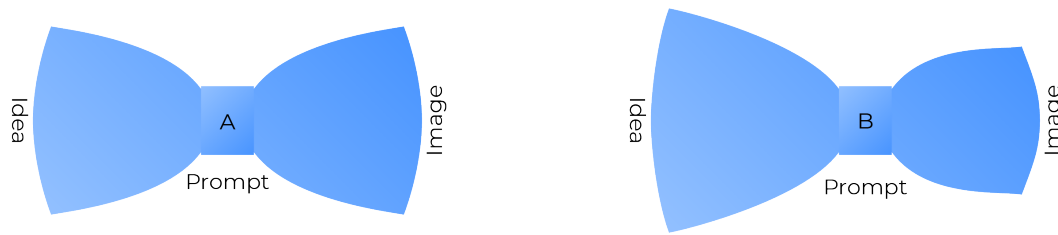


Figure 4: Visual representation of crooked bow tie effect. The image generation process can be seen as a bow tie. When used for inspiration (A), an idea converges into a prompt, from which an image is generated. Both sides of the bow tie roughly feature equal richness, just in different formats. When used for final designs (B), the bow tie becomes crooked on the left. Now, a very rich and complex concept must be converted into a prompt, from which it is very difficult to generate an equal amount of richness or accuracy.

elicit more elaborate feedback and suggestions [20]. People who are confronted with this technology for the first time, have little frame of reference, will likely feel that the machine actually “designed” something. Yet, it does not. It simply draws an image of a backpack. The next time, a different backpack will be generated, in random variation. No conscious decisions are made to create coherent form, function, and interaction. The human designer should not only be in the loop, but also *direct* the loop to ensure creative and well-informed decisions, while leveraging generative AI as a supportive tool for repetitive tasks such as making 100 drawings for inspiration.

The feedback on BrainFax has been predominantly positive. Despite initial difficulties, study participants as well as visitors of a recent faculty exposition shared their enthusiasm and saw potential for use in their own work. It is difficult to say how specific use cases will arise in different application fields, but with our exploration of generative AI in a designerly context, we have illustrated the potential strengths and weaknesses of process- and result-oriented approaches in a professional context: a process-oriented approach aims at opportunities for inspiration and collaboration. When the optimal visual quality and accuracy is deferred to different process stages, this leaves room for conceptual work on own interpretation, constructive discussion, and other collaborative activities. In short, AI is used to enhance the design process, not replace (even a part of) it. The result-oriented approach on the other hand resulted in struggles to generate sought-after desired images, often based on an existing concepts or specific functionalities. This is what we called the “crooked bowtie” effect, in which AI is essentially used to replace the design process, not enhance it. Interestingly, this was not found in previous work on visual co-creation tools that used existing images, but this perspective is not uncommon.

5.1 Guidelines for Collaborating with Generative AI

As illustrated, there are several challenges for using generative AI in creative collaboration. With the right implementation, however, there is potential to enhance and facilitate visual collaboration: “A great idea to use in collaborative processes, especially with clients who are not able to explain themselves visually through sketching”. Therefore, through a thematic analysis of both testing and co-creation sessions, we provide guidelines in the form of a workflow to integrate generative AI into creative collaborative work effectively, which

is displayed in Figure 5. The workflow consists of three repeating phases: (1) definition, (2) generation, and (3) reflection, which structure and balance individual generation and collective discussion activities. The definition phase establishes a common starting point and expectations for image generation. This includes a session goal and possible directions for image generation, i.e., what and why are we generating? This does not only provide guidance and coherence throughout image generation but also throughout reflection and analysis. Although specific styles, attributes, and keywords can be defined, we argue that these might make the scope too narrow and limit ad-hoc ideation and inspiration. The generation phase is used to make, share, and edit images for inspirational or communicative purposes. The use of the generated material is, therefore, two-sided. On the one hand, it can be used to be inspired and quickly explore multiple ideas individually. On the other hand, material that elicits new ideas or perspectives can immediately be used to communicate it to others. Although this can be done together, we argue that an individual approach to image generation allows for ad-hoc exploration of multiple perspectives without continuous deliberation. Finally, we recommend using a time limit to prevent endless diverging. The reflection phase is used to strategically organize and analyze generated material to facilitate discussion and value creation. Using the goals and directions set during the definition phase, satisfactory images can be selected, e.g., through voting. These images can then be analyzed for overlapping characteristics which essentially describe the design preferences of the team. This can be used to inform another ‘loop’ of the workflow, ad-hoc generations throughout the discussion, or other design activities. Therefore, the value of generative AI goes beyond the generated material itself, but provides knowledge for the team itself or other teams and stakeholders.

5.2 Limitations

There were limitations in this work that need to be addressed, but also provide directions for future work. First, all participants in this project worked at the same design agency and focus on interactive product design. Applications in other fields like fashion or interior design were not explored, which might have yielded different results due to different field-specific practices and ways to involve creative collaboration. Secondly, the BrainFax workflow was developed as a response to and in collaboration with participants

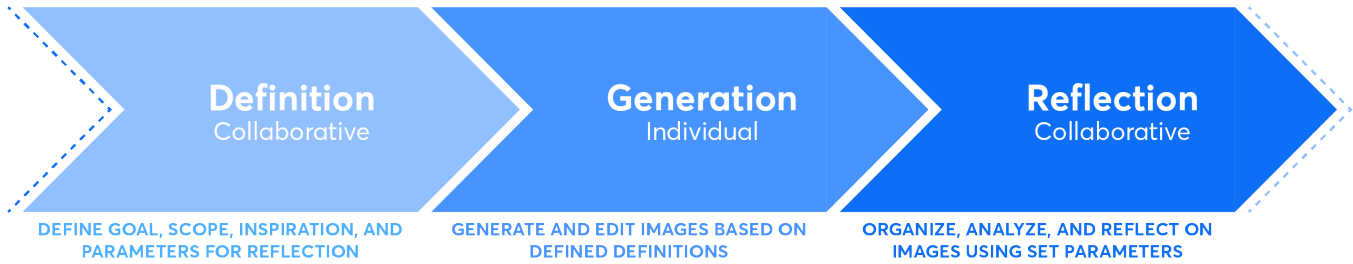


Figure 5: Visual representation of proposed workflow for collaborating with generative AI. The workflow consists of three repeating phases: definition, generation, and reflection, in chronological order. The definition phase provides a common starting point for image generation and editing. The reflection phase serves as a structured way of interpreting the generated material, guiding further iterations through the workflow. The dashed lines indicate that the workflow repeats itself. The definition and reflection phases are conducted collectively, whereas the generation phase is individual.

in the field but was not re-validated in actual use. Although our results, to some extent, support that the three-phase structure of definition, generation, and reflection is beneficial to implementing AI-generated images, the specific activities in each phase might need adjustment depending on team and application domain. Third, Stable Diffusion was used in this project because it is available as open-source software and usable even with consumer-level equipment. Using MidJourney or DALL-E 2 might have yielded different results in terms of quality and accuracy. In our experience, we found MidJourney to generate considerable higher-quality and more accurate results, but that comes at the cost of a much longer generation time, which is less suitable for the ad-hoc and dynamic nature of co-creative, collaborative design work. We found DALL-E 2 to be comparable to Stable Diffusion in terms of quality and speed, but it yields less diverse results and is less permissive in prompting. Additionally, in terms of AI capabilities, BrainFax was not fully featured. Due to technical constraints, in- and outpainting was not possible, for example. As these features can make the image editing process easier and more accurate, future work should include them and explore how they influence the collaborative expression of ideas.

Finally, we are aware of and acknowledge the ethical concerns surrounding data provenance for training deep learning models and copyright infringements by generative AI systems due to their training on publicly available material, often taking a liberal stance on fair-use regulation. This has implications on the use of generative AI in design work, especially in terms of copyright, which remains a gray area in this domain. Although it is unlikely for these diffusion models to return source material, certain artist or brand styles can be reproduced to be nearly identical, albeit using different subjects or composition. This essentially allows users to design with approximate style of someone else if their work was part of the training material. This further blurs the lines of ownership and copyright on AI generated material. But apart from *how* training data is collected and processed, *what* data is actually used raises additional ethical concerns in terms of bias [12, 25]. To iterate on an earlier point: better understanding the underlying technology and its limitations as well as further research on the implications of integrating generative AI in creative workflows is clearly necessary. Related work on machine learning often favors benefits for

research and industry, thereby overlooking the potential negative societal implications [2]. In the context of this research, generated material plays a special intermediary role, which, in our view, has less potential for harm given careful embedding in design activities.

6 CONCLUSION

Visual communication is essential in complex collaborative design contexts, but an increasing lack of hands-on visualization skills discourages us to work that way. In this paper we explored how generative AI can be used to support creative collaboration in terms of inspiration and communication. To investigate this application, we developed BrainFax, a multi-user, process-oriented visual co-creation tool using AI image generation for collaborative conceptual design activities. With BrainFax, users can individually generate images using AI, share them to a collective digital whiteboard, and iterate on images from others using simple sketches. Given the feedback from co-creation sessions with designers, it was evident that methodological guidelines are required to direct the generation process and productively integrate the generated material. Through a co-creation session with designers in the field, we developed guidelines for integrating generative AI into creative processes, consisting of repeating definition, generation, and reflection phases. This provided users with handles for creating a common goal and method for analysis, but also suggests directions for future research to (1) complete this initial frame of guidelines and (2) validate it with a broader range of creative professionals and collaborative stakeholders. We close with a discussion touching on the “crooked bow tie” effect that connects aesthetics and richness with the fundamental mechanisms of generative AI, and on the risks of misinterpreting capabilities of AI-supported technology—how it can enhance design, not replace it.

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